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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - MAY'2025

B.E – COMPUTER SCIENCE AND ENGINEERING

20CS2E48 – SOFTWARE TESTING

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1** What is a test case?
 - a) A list of all the bugs in the software
 - b) A document that describes how a test should be executed
 - c) A report of testing results
 - d) The final version of the software

2. **CO1** What is the difference between verification and validation?
 - a) Verification checks if the product is built correctly, validation checks if the product is correct
 - b) Verification is testing, validation is documentation
 - c) Verification is about testing the product in the real world, validation is done by developers
 - d) There is no difference

3. **CO2** System testing is a _____.
 - a) Grey Box Testing b) White Box Testing
 - c) Black Box Testing d) Both (b) and (c)

4. **CO2** What is the first level of testing usually performed on software?
 - a) Integration Testing b) System Testing
 - c) Acceptance Testing d) Unit Testing

5. **CO3** Which of the following is a primary objective of reporting in software testing?
- a) Minimizing coding errors
 - b) Documenting team communication styles
 - c) Tracking defects and monitoring progress
 - d) Designing user interfaces

PART - B

(5x 3 = 15 Marks)

6. **CO1** How do manual testing and automated testing differ?
7. **CO1** What is the importance of Test Requirement Specification in the software testing process?
8. **CO2** How do you identify bottlenecks during performance testing?
9. **CO2** Define non-functional testing and list at least five types with examples.
10. **CO3** In what situations would an informal review be more effective than a formal review?

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) Compare and contrast different types of software testing (unit testing, integration testing, system testing, acceptance testing, etc.). (5)
- CO1** ii) Differentiate between test cases, test scenarios, and test scripts with examples. (5)
12. **CO1** i) Describe the stages of the Testing Life Cycle (TLC) and explain how each contributes to software quality. (5)
- CO1** ii) How does early involvement of testing in SDLC impact the overall project outcome? (5)
13. **CO2** i) Compare white box testing and black box testing in terms of techniques, benefits, and use cases. (5)
- CO2** ii) Mention the key steps in creating automated test procedures for a software system? (5)

14. **CO2** i) Write the importance of security testing and explain the consequences of security breaches, also write the various areas which have to be focused on during security testing. (5)
- CO2** ii) State the need for integration testing in procedural code. (5)
15. **CO2** i) Describe the top-down and bottom-up approaches in integration testing discuss about the merits and limitation of these approaches (5)
- CO2** ii) Suppose you are developing an online system for a specific vendor of the electronic equipment with all the necessary features to run the Shop. Write down a detailed test plan by including the necessary components. (5)
16. **CO3** i) Enumerate the various types of reviews conducted in software testing. Provide a brief explanation of each type along with suitable examples illustrating their application. (5)
- CO3** ii) How does the review plan contribute to early detection of defects and project control? (5)
17. **CO3** i) Critically compare reporting review results in agile vs. traditional development environments. What challenges might arise in each, and how would you address them? (5)
- CO3** ii) How can review processes contribute to the evaluation of software quality? (5)
